Hamza Khan

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Professional Summary

Machine Learning Engineer with experience developing and deploying AI-driven applications and deep learning models in production environments. Skilled in Python development, REST API services, CI/CD automation, and containerized deployments using Docker and AWS. Experienced in building scalable ML pipelines, evaluating model performance, and integrating AI solutions into production systems within Agile development environments.

Technical Skills

Programming: Python, JavaScript, Java, SQL, Bash
AI / Machine Learning: PyTorch, TensorFlow, Scikit-learn, CNN, LSTM, model evaluation
Generative AI: Prompt engineering, LLM integration, HuggingFace, OpenAI APIs
Backend Development: Flask, REST APIs, Microservices
Cloud & DevOps: AWS (EC2, S3, IAM, CloudWatch, ALB), Docker, Kubernetes, Jenkins, GitHub Actions, CI/CD
Testing & Monitoring: Automated testing pipelines, performance evaluation, system monitoring
Databases: PostgreSQL, MySQL
Tools: Git, Linux, Jupyter Notebook, Agile/Scrum

Education

North Carolina A&T State University — Greensboro, NC
Ph.D. in Computer Science (In Progress), GPA: 3.90 — *2024 – Present*

- Research focus: AI-enhanced Cyber-Physical Systems, deep learning for surveillance violence detection, and system reliability
- Assist faculty in mentoring students and supporting coursework related to machine learning and software engineering

North Carolina A&T State University — Greensboro, NC
M.S. in Computer Science, GPA: 3.88/4.00 — *Aug 2022 – Oct 2023*

University of Haripur — Haripur, Pakistan
B.S. in Software Engineering, GPA: 3.90/4.00 — *Aug 2016 – Oct 2020*

Experience

Lead Developer (Technical Project Manager) — Feb 2025 – Present
EzyTeams

- Led development and delivery of SaaS-based collaboration tools working closely with cross-functional engineering teams
- Oversaw feature implementation, backlog prioritization, and release planning using Jira and Trello
- Supported engineering teams in resolving technical blockers and coordinating end-to-end delivery of product features
- Facilitated stakeholder communication through product demos, progress updates, and roadmap discussions
- Maintained development timelines to ensure alignment between product goals and engineering execution

Graduate Research Assistant — Aug 2022 – Present
North Carolina A&T State University — *Greensboro, NC*

- Designed and developed Python-based machine learning pipelines for violence detection in surveillance videos using deep learning models
- Developed automated evaluation pipelines for deep learning models including performance benchmarking and inference monitoring.
- Developed Python-based utilities to integrate LLMs into AI workflows, implementing prompt templates, automated response evaluation, and REST API services for scalable inference.

- Designed automated evaluation pipelines for AI models, benchmarking inference performance and validating outputs across multiple datasets to improve reliability and deployment readiness.

DevOps Engineer Sep 2022 – Nov 2024

Byte Forge Information Technology Services *Dubai, UAE*

- Developed automation pipelines for application builds, testing, and deployment using Jenkins, GitHub Actions, and GitLab CI
- Provisioned scalable infrastructure using Terraform and CloudFormation on AWS cloud environments
- Deployed containerized Python applications using Docker and Kubernetes ensuring high availability
- Improved operational stability by automating monitoring, logging, and deployment workflows
- Worked closely with development teams to debug production issues and optimize system performance
- Automated CI/CD pipelines for containerized microservices supporting machine learning services.

Full Stack Developer May 2021 – May 2022

Pakistan Professional Development Academy *Haripur, Pakistan*

- Built and launched the Inspire London College website using Laravel and JavaScript.
- Developed and deployed web applications using Laravel and React with REST API integrations.

Projects

Deep Learning Model Deployment Pipeline

- Designed and implemented an end-to-end ML deployment pipeline for training, packaging, and serving models
- Built Python-based inference APIs using Flask and deployed containerized services using Docker on AWS EC2
- Implemented CI/CD pipelines using GitHub Actions to automate model deployment and updates

Violence Detection in Surveillance Videos

- Developed CNN and LSTM-based models for temporal violence detection in industrial surveillance environments.
- Designed modular processing pipelines optimized for real-time video stream analysis and deterministic system behavior.
- Implemented structured validation workflows to evaluate performance, reliability, and false-positive reduction.
- Focused on deployment considerations for resource-constrained and latency-sensitive systems.
- Deployed real-time inference APIs using Flask, Docker, and AWS EC2

Human-Centric Security in AI-Enhanced CPS

- Analyzed human behavioral vulnerabilities in cyber-physical systems
- Proposed mitigation strategies for AI-integrated CPS environments
- Examined interactions between software control logic and physical system components within Cyber-Physical Systems (CPS) environments.

Embedded Industrial Motor Control System (MCU-Based)

- Designed and implemented a bare-metal embedded system in C on STM32/AVR microcontroller.
- Developed a finite state machine (FSM) to control motor states including Idle, Running, and Fault.
- Implemented external interrupt handling for input controls and timer-based fault monitoring.
- Integrated UART-based logging for real-time system diagnostics.
- Created validation test cases and documented state transition behavior.

Certifications

AWS Certified Cloud Practitioner (In Progress)

IBM DevOps (Coursera)